Learning Day 4 — Updated Simple Visual Notes

Outlier Detection

Professional theory notes rewritten in a **very simple, visual, beginner-friendly style**. This version updates the previous PDF with clearer explanations of **Point, LOF, LoOP, ABOD, Isolation Forest, PCA, Jaccard, CBLOF, and Autoencoder**.

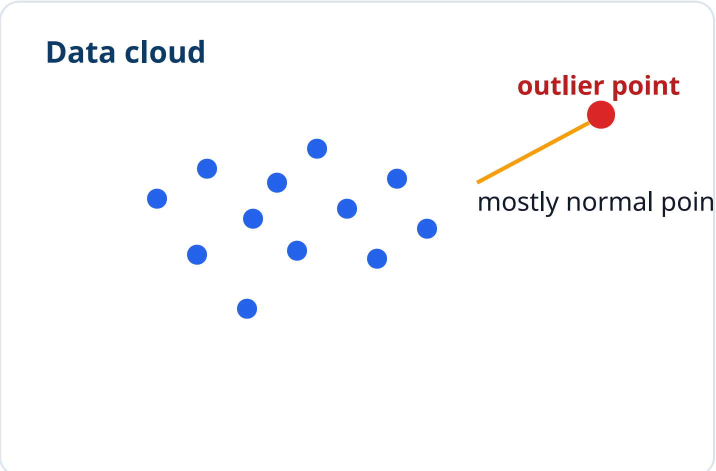
First idea: what is a “point”?

In machine learning, a **point** means **one row in the dataset**.

Row / Point	Temperature	Pressure	Vibration	Meaning
Point 1	75	30	0.3	normal
Point 2	78	32	0.4	normal
Point 3	74	29	0.2	normal
Point 4	80	31	0.5	normal
Point 5	**180**	**95**	**8.9**	**suspicious**

> **Simple meaning:** when we say “this point is an outlier”, we mean “this row is unusual compared to the other rows”.

Normal vs Outlier



Memory story

Imagine each row is a **child in a playground**.

- Most children stand in groups.
- One child may stand far away.
- One tiny strange group may stand in a corner.

Many points stay together. One unusual point lies far away.

- That is what anomaly methods try to find.

Main story for this PDF: every algorithm will be explained using this playground / point / row idea.

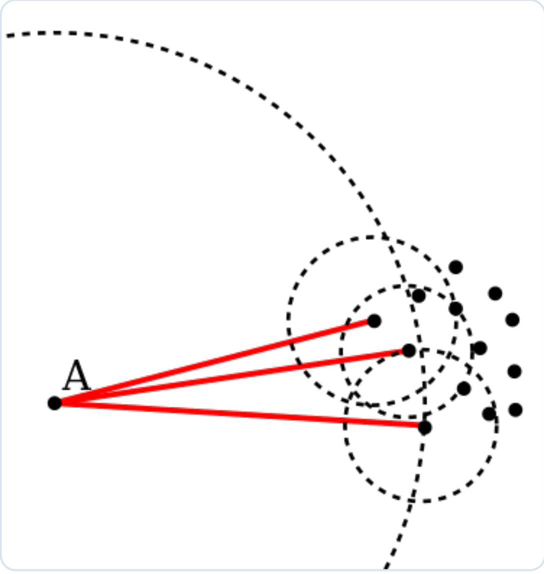
1) LOF — Local Outlier Factor

LOF asks: “Is this row lonely compared to nearby rows?”

LOF looks at the **local neighborhood**. It does not simply ask whether a point is far from everyone in the whole dataset. Instead, it asks whether a point is less crowded than the points around it.

- Find nearby neighbors.
- Check how crowded that area is.
- If the point is much less crowded than its neighbors, it becomes suspicious.

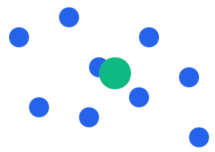
Memory trick: LOF = **Lonely compared to nearby points**.



Actual intuition image for LOF / local neighborhood idea.

Dense local group

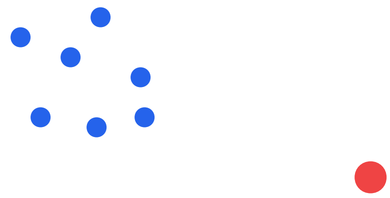
Point inside local crowd



Many close neighbors → likely normal

Locally lonely point

Point with few nearby neighbors



LOF says: locally lonely -

2) LoOP — Local Outlier Probability

> **LoOP asks:** “How likely is this locally lonely point an outlier?”

LoOP is close to LOF, but it gives a score that is easier to explain. Instead of only saying a point is suspicious, it gives a **probability-like score** between **0 and 1**.

- Near **0** → probably normal
- Near **1** → probably an outlier

> **Memory trick:**
> LoOP = **LOF with probability**.

LoOP score scale

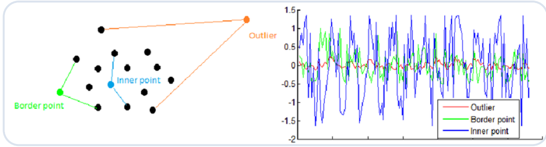
LoOP makes the output easier to read for beginners and business users.

3) ABOD — Angle-Based Outlier Detection

> **ABOD asks:** “Is this point surrounded from many directions, or are the other points mostly on one side?”

ABOD uses **angles** instead of mainly using distance. The idea is simple:

- If a point is **inside** a group, other points appear in many directions.
- If a point is **outside** the group, most other points appear in one broad direction.

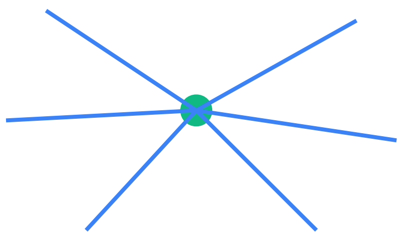


Actual intuition image for ABOD showing inner point, border point, and outlier behavior.

> **Memory trick:** ABOD = **Am I surrounded from many angles?**

Point inside cluster

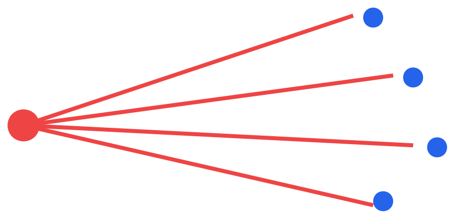
Many directions



Likely normal

Point outside cluster

Mostly one direction



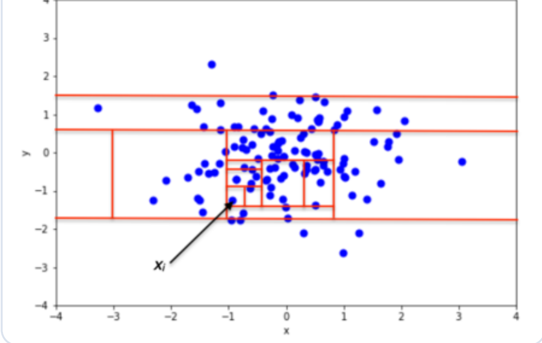
Likely outlier

4) Isolation Forest

> **Isolation Forest asks:** “How quickly can I separate this row from the other rows?”

If a row is very different, it is easy to isolate using random splits.

Point	Temperature

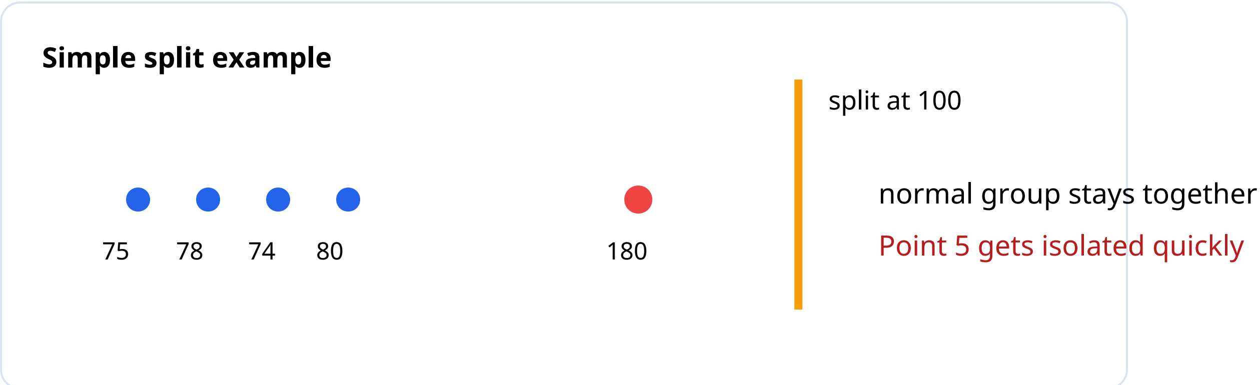


Actual intuition image for Isolation Forest.

Point 1	75
Point 2	78
Point 3	74
Point 4	80
Point 5	**180**

Example split: Temperature < 100? Then Points 1–4 go to one side, and Point 5 is isolated immediately.

Memory trick: Isolation Forest = **Weird rows are easy to separate**.



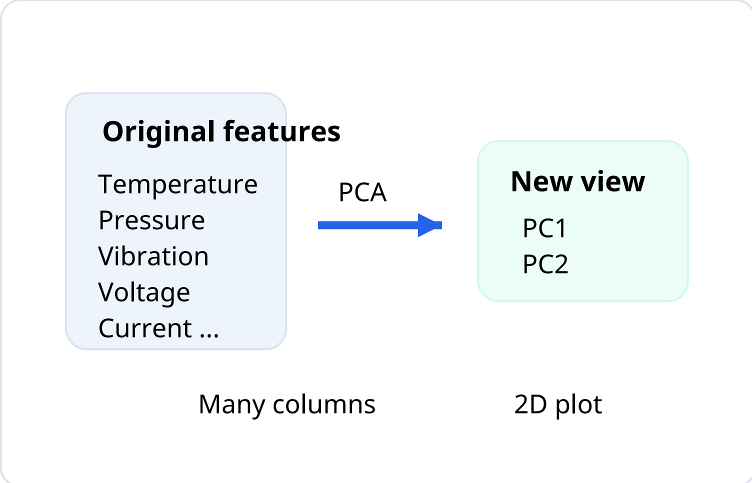
5) PCA — Principal Component Analysis

> **PCA asks:** “Can I turn many columns into a simple 2D picture?”

PCA is mainly used for **simplification and visualization**.

Suppose the dataset has many columns:

> Temperature, Pressure, Vibration, Sound, Voltage, Current, RPM, Humidity, Oil level, ...



Humans cannot easily see 10 or 20 dimensions at once. PCA creates two new summary axes, often called **PC1** and **PC2**, so we can draw the rows as a scatter plot.

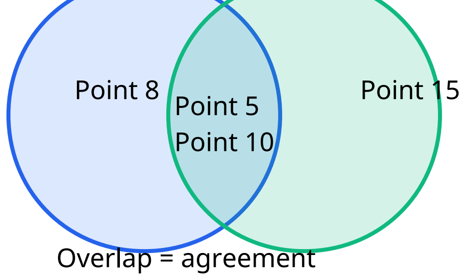
> **Memory trick:**
> PCA = **Turn many columns into a simple picture**.

6) Jaccard Coefficient

> **Jaccard asks:** “How much do two methods agree?”

Suppose two methods each return a list of outlier rows.

- **Isolation Forest**: Point 5, Point 8, Point 10
- **LOF**: Point 5, Point 10, Point 15



Shared points = **Point 5, Point 10**

All unique points = **Point 5, Point 8, Point 10, Point 15**

Jaccard = 2 / 4 = 0.50

> **Memory trick:**
> Jaccard = **How much same?**

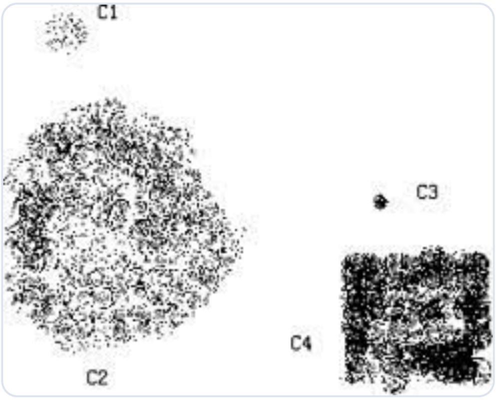
7) CBLOF — Cluster-Based Local Outlier Factor

> **CBLOF asks:** “Is this point in a tiny strange group, or far from a cluster center?”

A **cluster** is a group of similar points. CBLOF uses cluster information.

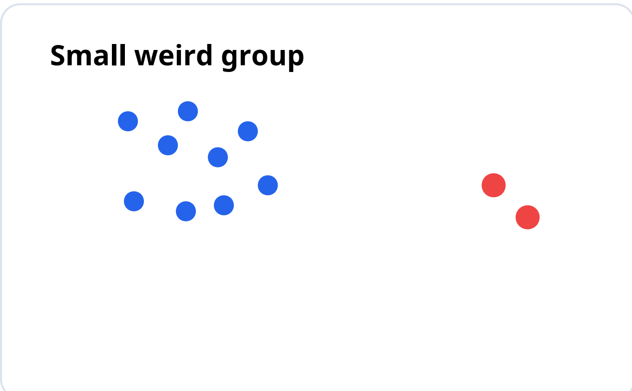
- Big normal clusters are usually less suspicious.
- Tiny strange clusters can be suspicious.
- A point far from its cluster center can also be suspicious.

> **Memory trick:** CBLOF = **Small strange group or far from cluster center**.

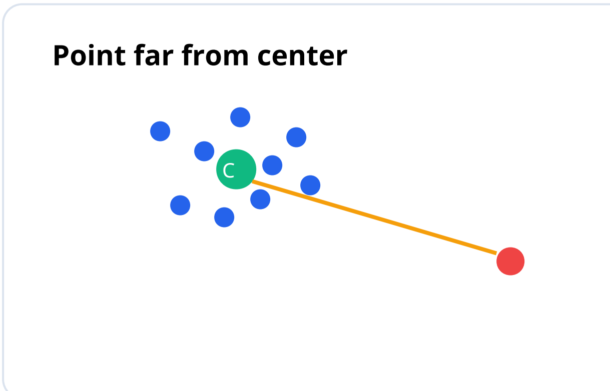


Actual intuition image for CBLOF-style cluster logic.

Tiny strange cluster



Far from cluster center

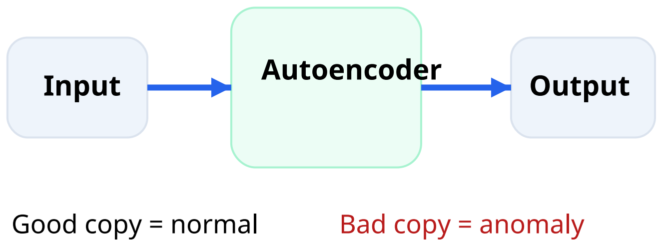


8) Autoencoder

Autoencoder asks: “Can I copy this row well?”

An autoencoder learns the pattern of **normal rows**. Then it tries to reconstruct a row.

- If the row is normal, the copy is good → **small reconstruction error**.
- If the row is strange, the copy is poor → **large reconstruction error**.



Memory trick: Autoencoder = **Learns normal rows and fails to copy strange rows**.

Normal row

	Temperature	Pressure	Vibration
Input	75	30	0.3
Output	76	31	0.4

Small difference → small error → likely normal

Suspicious row

	Temperature	Pressure	Vibration
Input	180	95	8.9
Output	79	32	0.5

Large difference → large reconstruction error → likely anomaly

Final Simple Comparison Table

Method	Very simple question	Memory trick
LOF	Is this row lonely compared to nearby rows?	

		Lonely compared to neighbors
LoOP	How likely is this lonely row an outlier?	LOF with probability
ABOD	Is this row surrounded from many directions?	Many angles = normal
Isolation Forest	Can this row be separated quickly?	Easy to isolate = suspicious
PCA	Can we draw many columns as a 2D picture?	Shrink data into a picture
Jaccard	Do two methods agree on the same outliers?	How much same?
CBLOF	Is this row in a tiny group or far from a cluster center?	Small group / far center
Autoencoder	Can the model copy this row well?	Hard to copy = anomaly

One final playground story

- **LOF**: Is the child lonely compared to nearby children?
- **LoOP**: What is the probability that this lonely child is unusual?
- **ABOD**: Are children around this child in many directions?
- **Isolation Forest**: Can this child be separated quickly with fences?
- **PCA**: Can we draw the whole playground as a simple 2D map?
- **Jaccard**: Do two teachers agree on which children are strange?
- **CBLOF**: Is the child in a tiny strange group?
- **Autoencoder**: Can a model trained on normal children copy this child's behavior?

Updated PDF version created from the previous professional notes, now with simpler explanations, clearer diagrams, and actual intuition images.